SoK: Agent Communication Side Channels Feedback Paths and Executed Cover

***Abstract*—Agent traffic defenses must protect information as it moves between services and constrain what cover traffic executes. We systematize fifteen Agent studies, forty-four settings and twenty-two traffic-analysis anchors to connect observable signals, access rights, defense placement and execution outcomes. Two findings organize the evidence. First, tool destinations and model connections expose different representations of a task: evaluating a tool-side defense requires checking the other views available to its attacker. Second, executing cover introduces task and resource obligations that transmission overhead cannot capture. In published research-decoy evidence, the reported privacy gain and original-task utility belong to different interventions. A deployment map and transfer matrix show where established traffic-analysis guidance applies and where Agent execution adds these obligations. A controlled MCP study illustrates joint reporting of recognition, task completion, business state and cost; its deliberately adverse writes exercise the distinction between target success and collateral effects. A separate forwarding pilot leaves feedback's incremental traffic effect unresolved. We provide an evaluation procedure and a source-dependent research agenda. Published results, preprint-dependent measurements and controlled examples retain distinct evidential roles.**

1. Introduction

An Agent receives the instruction *process the next support ticket*. It reads the ticket, checks an order and approval, consults a model, and submits a refund. Encryption hides message contents from a network observer while contacted services, sizes and timing remain visible. The tool's answer then enters a model request, creating another observable representation. Adding cover raises a second question: does it end at a proxy, execute a read, or change business state? Figure 1 follows information and actions through this example.

Defense placement determines both remaining exposure and permitted execution effects. A tool-side intervention covers a particular interface. Its evaluation must also account for other interfaces available to the attacker and for any work the intervention causes. This SoK connects fifteen direct studies, forty-four settings and twenty-two established traffic-analysis anchors around these two requirements.

Follow information across services. WRA recovers research queries from tool destinations, while ANT and the cloud benchmark recognize task labels on model connections [2], [9], [4]. A contacted tool has yet another view: its addressed arguments. These observations change the unit of protection from a single transformed trace to a declared set of accessible interfaces. The field map identifies those interfaces, the rights needed to observe them and the evidence for each signal. Our forwarding pilot separately tests the incremental effect of supplementary feedback; its corrected results leave that causal question open.

Follow cover into execution. Transport cover can terminate without invoking application work; an Agent decoy may execute queries or update state. WRA's published research-decoy comparison reduces query recovery, while its original-task utility scores concern a different visibility intervention [2]. The practical consequence is to measure privacy, original-task outcomes and resource use under the same cover policy. We illustrate this record with six paired policies and two fixed-weight planners over MCP Streamable HTTP. Deliberately adverse writes test whether a target-task predicate captures collateral changes; read, discarded and isolated cover provide the accompanying controls. This is an evaluation example, not a finding of harmful writes in WRA.

Website fingerprinting (WF) and machine-learning security already establish realistic populations, segmentation, transfer and adaptive evaluation [15], [45], [32], [16], [52], [19]. Our contribution is their execution-aware systematization: a deployment and transfer map; two source-grounded syntheses connecting defense placement to exposure and task effects; and a worked procedure for reporting both together. The maps distinguish transit, recipient and active access instead of combining their scores. Source-dependency analysis identifies which comparisons survive without preprints. Together, these artifacts turn a defense's placement into concrete observation and outcome requirements.

2. Scope and Evidence Construction

The evidence map connects Agent execution to an observed communication signal, a hidden target and the rights needed to observe it.

2.1. Inclusion and retrieval

An Agent uses an LLM to select actions, with external feedback able to affect later action or termination. Client, tool, model and inter-agent interfaces are eligible. Transit observers obtain traffic metadata; contacted recipients see

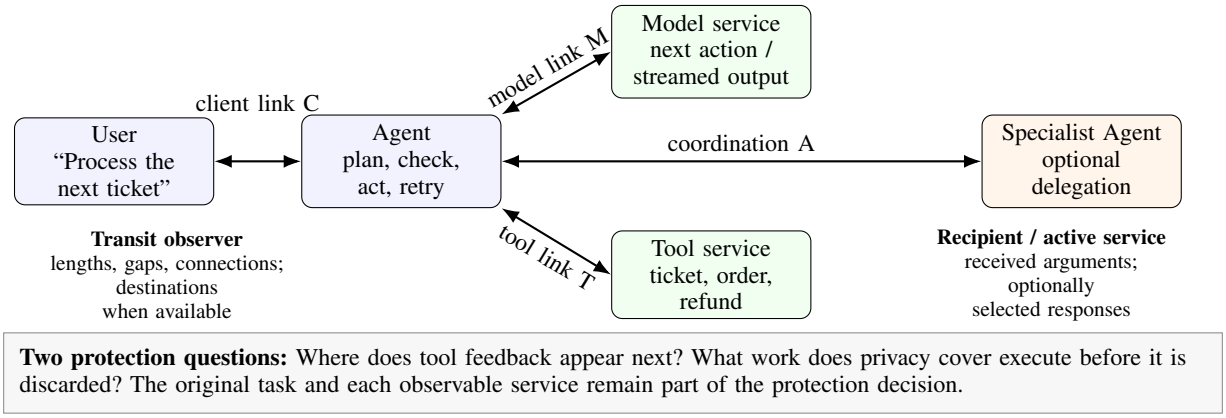


Figure 1. Running support-ticket example and observation rights. A tool result returns to the Agent and can enter the next model request. Each encrypted link exposes its own traffic shape; a recipient additionally sees the request addressed to it. Concurrent tickets may share a connection on either link. The arrows represent application communication, not a one-to-one mapping from tool events to packets.

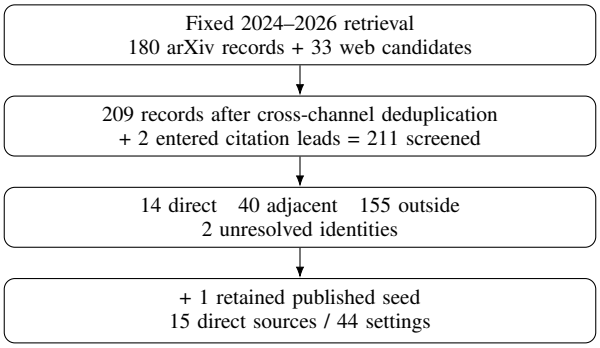


Figure 2. Executed coverage audit. Counts are deduplicated work/lead records; two identities remain unresolved. Mechanism anchors use a separate inclusion rule.

their addressed requests; active auditors also choose probes or challenges. Negative results qualify: PrivacyTrace's metadata/order condition remains included despite scoring below its prior reference [35]. Whole semantic trajectories or browser instrumentation alone enter the adjacent layer, while deliberate covert messages and prompt-injection defenses without communication-side inference remain contextual related work [36], [26].

The fixed eligibility window covers first public submissions from 1 January 2024 through 9 November 2026. Twelve topic queries, a native arXiv query and inclusion rules preceded the retrieval pass; title/publication checks and one-step citation follow-ups completed it. The arXiv query returned 180 records. Public-web queries supplied 33 scholarly candidates across 101 displayed URLs; deduplication yielded 209 records. Two citation leads brought screening to 211: fourteen direct, forty adjacent, 155 outside scope and two unresolved identities. The eligible CHeaT (Cloak, Honey, Trap) seed adds the fifteenth direct source (Figure 2). The SSH honeypot entered through citation tracing, illustrating the value of combining retrieval routes. Exact queries and dispositions are in the supplement.

Twenty-two separately selected mechanism anchors comprise sixteen published communication studies and six serving studies. They supply the established relations in the transfer matrix; the fifteen direct studies supply the Agent application evidence. These layers retain separate inclusion records and denominators. The anchor set is selected for mechanism correspondence, rather than a census of classical traffic analysis.

2.2. Extraction and source dependencies

Forty-four setting records bind source version and locator to access, features, learner, target, population, split, intervention, result and cost. Labels, execution outcomes, failure treatment and policy context are recorded alongside them. WRA and CHeaT are checked against official USENIX texts. One HTTP study has a verified ACM workshop identity, while its detailed measurements come from the extended arXiv version [49], [50]. Twelve other works lack independently verified formal publication identities at the cutoff.

Text reading, displayed-number reconstruction, selected-code inspection and data recomputation have separate entries in the extraction sheet. For example, WRA's decoy and visibility conditions are checked in the published text; OHTTP refitting is checked in selected code. Each check supports its matched premise. Section 9.1 reports what remains when sources are removed. Extraction is author-performed with AI assistance. A coding-manual pilot compares the released author coding with two non-blind AI passes from the same model. Structural fields agree on all forty-four rows, while execution-outcome definitions produce substantial disagreement. Supplement S4.1 gives the original encodings, descriptive agreement and source adjudication; this is not independent human validation.

For the descriptive summary, every setting receives an access class and intervention purpose. Protected-condition adaptation and original-task utility are counted within the four query/workflow privacy settings. Three identity-

obfuscation settings and two adversarial-Agent blocking settings have different objectives; thirty-five settings contain no protection comparison. A located item identifies evidence in the retained record. An unlocated item leaves that question open. The supplementary reporting ledger lists the assignments and source locators.

2.3. Relation to security surveys and Agent defenses

Broader surveys organize LLM and Agent threats by lifecycle, attack surface and defensive role [53], [54], [55]. This SoK asks three communication-specific questions: which service representation reaches which observer, whether an intervention covers the later feedback path, and what business work its cover can trigger. The maps connect each answer to a measured setting and its evidence strength. The supplementary deployment section compares these surveys and WF systematizations.

Information-flow and privilege controls address closely related execution boundaries. CaMeL tracks dependencies and explicitly discusses tool-call and timing side channels [57]; FIDES enforces confidentiality/integrity policies and selectively exposes tool information [58]; Progent constrains tool calls and arguments [59]. SPILLage distinguishes content and behavioral oversharing by browser Agents [27]. Our maps add the communication-observer question: which metadata remain visible under a particular deployment and intervention? They do not replace these policy mechanisms. Likewise, τ-bench and AgentDojo already evaluate environment outcomes [60], [61]. The cover example applies state-aware evaluation to matched traffic measurements; state checking itself is established practice.

3. Reading the Evidence Map

The same ticket can expose different facts at the network link, the tool recipient and the model service.

3.1. From private task to observed execution

An execution maps a hidden task or state through an agent policy and environment into communication, then into an observer's features:

$$\text{hidden task/state} \longrightarrow \text{agent execution} \longrightarrow \text{observed communication} \longrightarrow \text{inference}.$$

Different cuts through this chain expose different objects. A client link can reveal response mode or hosted-agent identity. Tool egress exposes contacted services and request/response shape. The model link carries prompts, tool feedback and model responses. Inter-agent links expose delegation and coordination. Figure 1 places these observations on the feedback path.

3.2. What the inferred label means

The ticket example separates four labels. *Intent* is the requested objective or assigned task family. *Execution* is the recorded action sequence. *Success* states whether that sequence and its final state meet the task's criterion. *Authorization* states whether the action is permitted for the principal under the applicable policy. A failed refund task can still contain a refund attempt; a successful refund can still be unauthorized. An evaluation should bind its target to the corresponding reference instead of letting one label stand for all four.

Exposure is relative to both the target and the observer. The same execution may expose intended-task information on M and actual contacted services on T. Initial instructions, tool feedback and later decisions all contribute to the complete M trace. Assigning a task-family label to that trace measures intention recognition; attributing the signal to a particular feedback step requires a further controlled comparison.

3.3. What each record specifies

Six fields give a score its meaning: the *interface and visible representation*; *observer access*; *prior information* supplied by the evaluator; *target*; *population and base rate*; and *protected training access*. These fields define the comparison record. The matrix records the fields once; the reporting checklist in Section 8 turns them into entries for future studies.

Access determines what a feature can mean. A server can inspect decrypted HTTP/2 properties that a transit observer cannot. The content-aware MCP study's "metadata" includes argument hashes, and its graph uses content matches [42, Secs. 4.1–4.2]. Its representation therefore requires more than packet metadata. Logs and browser instrumentation can provide evaluation labels without becoming attacker inputs.

ANT's supplied stage boundary is prior information; its 2,101 stages and 47 labels define the decision population; the primitive is the target [9]. In the live OHTTP comparison, an unchanged random-forest (RF) family refits on protected data, so it is an adapted attacker [3]. These examples show how to read access, target, population, supplied boundaries and training conditions in sequence. A source setting that needs an unrepresented field would require revising the map.

3.4. Deployment determines who can see each signal

Table 1 maps common deployments to observation rights and the corresponding source evidence. A visible model link is an access condition, not a property granted to every attacker. Captured packets, TLS records, HTTP streams, server-sent events (SSE) and tool calls are different units. Segmentation, buffering and retransmission change their correspondence. Supplement S7 gives the protocol-level map.

Two simultaneous tickets may use separate connections or share an encrypted HTTP/2 connection. In the latter case, a transit observer sees combined packets while stream

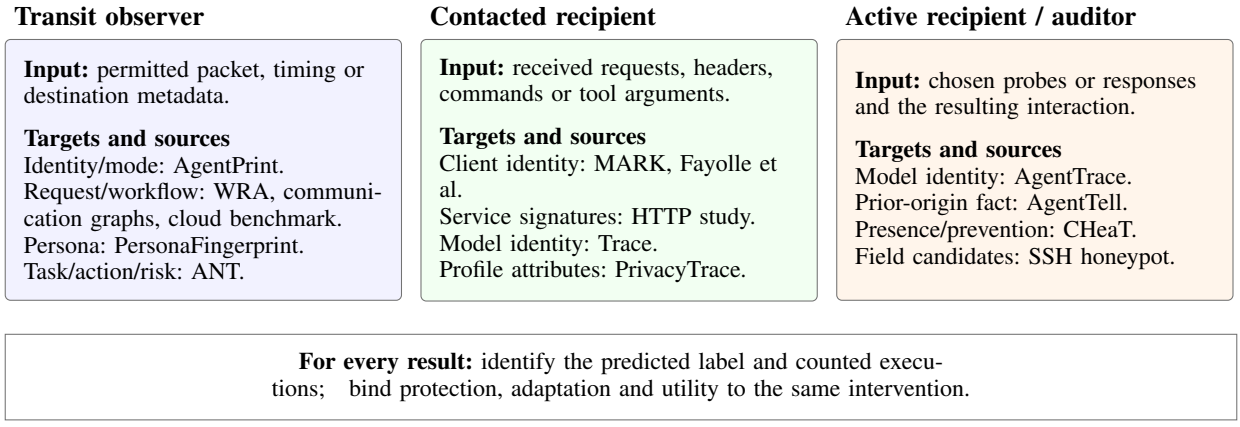


Figure 3. Evidence space by observation rights. The fifteen studies are placed by the mapped settings, not ranked by accuracy or access strength. Some offer richer additional views; Tables 3–4 identify those conditions and their metrics. Known-service signatures and field candidates are descriptive endpoints, not held-out recognition rates.

identifiers remain encrypted. A gateway that buffers SSE output changes the pattern again. The connection and flush policy therefore determine the measured trace.

The same reasoning explains defense placement. Padding a tool message before a gateway does not specify the gateway's output. Local MCP over `stdio` removes that network tool leg, while remote model requests may remain. Shared key/value (KV) cache state creates an additional timing interface for a querying tenant. The map therefore records the transport, observation layer, terminating parties and shared state alongside each result. Section 7.2 compares real HTTP/2 concurrency with paired feedback and shaping interventions.

4. Classical Conditions in Agent Execution

Established traffic-analysis results carry conditions on access, observation units, workloads and interventions. Table 2 connects those conditions to the Agent evidence. Its central distinction is between an evaluation principle that carries over and an empirical performance result that requires a matched Agent test.

4.1. Reading the transfer matrix

Open-world prevalence, false alarms, adaptive attacks and workload shift are established concerns [15], [46], [32], [16], [52], [19]. Localization also has direct precedents: Wang and Goldberg study traffic splitting without reliable page boundaries; BAPM and ARES address mixed multi-tab traces [45], [43], [44]. WF is neither inherently deterministic nor restricted to one link. Here these issues interact with feedback across independently controlled services and cover executed as application work.

Retained means that an established evaluation obligation still applies; *extended* identifies an additional Agent execution condition; *access mismatch* prevents importing a result under the proposed observer; and *untested* means that the mapped evidence leaves the specific performance claim open. These relationships are separate from the number and publication status of supporting sources. Distributed tracing illustrates access mismatch: WAP5 and vPath use host instrumentation [30], [31], which can label an experiment without becoming a passive attacker's input.

Five questions locate the resulting evidence: signal mechanism, supplied-unit recognition, named deployment transfer, stream localization and protection. They are distinct tests. A reported score binds a target to an observation and a tested deployment; Tables 3–4 preserve those bindings across the field.

4.2. Task boundaries and execution changes

AgentPrint supplies browser interactions, WRA research traces, the communication-graph study workflows, the cloud benchmark execution timestamps, and ANT runtime-defined stages [1], [2], [3], [4], [9]. ANT's 47-group primitive test scores 0.355–0.462 accuracy on 2,101 supplied segments. Its episode split separates parent executions, while the supplied boundaries define a recognition test. Event localization additionally requires recall and false events on proposed intervals.

PersonaFingerprint keeps one session's windows in one partition [5]. A neighboring published study classifies non-overlapping 5,000-packet windows without page markers, primarily testing human browsing with Agent-generated training traces [41]. PINGPONG and voice-invocation work demonstrate detection against background traffic [33], [22].

Transfer names what changes while the target remains fixed: wording, session, composition, topology, scaffold or model. Reported composition, topology and scaffold losses concern different axes (Section 5.3). Simultaneous Agent tasks and model waits also change how tool bursts relate to complete executions. Our live comparison separates supplied-window inference from localization and retains physical connection identities.

TABLE 1. DEPLOYMENT AND ACCESS MAP. THE CONFIGURATIONS ARE ANALYST-DEFINED; CITED SETTINGS TEST THE LISTED VIEW, NOT EVERY COMPONENT OF THE DEPLOYMENT. A GATEWAY'S PLAINTEXT ACCESS REQUIRES TLS TERMINATION.

Deployment	Observer and permitted view	Existing evidence for that view	Protection consequence
Enterprise egress	A transit monitor sees routed model/tool flows, sizes, times and any exposed destination association.	WRA: supplied research-domain traces; ANT and cloud benchmark: model/combined traffic [2], [9], [4].	Specify which services cross the monitored egress. A union of flows is available only when the monitor can observe them.
Shared gateway or relay	Transit sees multiplexed gateway output. A terminating gateway can additionally read addressed requests.	Live OHTTP: passive-link inference under stated relay trust [3]; PrivacyTrace: semantic recipient/gateway access [35].	Separate relay association protection from content trust. Connection sharing changes the observation unit.
Local model, remote tools	An external observer sees remote tool traffic; the local model call has no corresponding external network leg.	WRA's destination-recovery mechanism is relevant to tool egress [2]; whole-deployment protection is untested in the mapped comparison.	Evaluate tool and downstream calls. Do not import a remote-model attack score into an absent network leg.
Cloud model, remote tools	An authorized vantage may see either or both encrypted legs. A contacted tool sees its own arguments.	ANT/cloud benchmark test model traffic [9], [4]; PrivacyTrace tests recipient-visible representations [35].	Check permitted model and tool views after the intervention; recipients and passive monitors need different protections.

5. Two Execution Dependencies for Protection

The field map connects two properties of Agent execution to defense design: information passes between services, and cover can become application work. Table 5 states the observation conditions under which particular signals matter.

5.1. Feedback joins separately observable services

The support-ticket execution exposes a selected tool destination, arguments received by that tool, a later model request and the model's response. WRA recovers query meaning from contacted domains under its name-association and supplied-trace conditions [2]. ANT, a network-traffic benchmark for Agent behavior, and the cloud benchmark measure task recognition from model traffic [9], [4]. The cloud benchmark reports ten-class balanced accuracy .853/.860 for graph/star coordination with all 247 features, and .540/.587 when size features are excluded and 100 of the 134 remaining features are selected. Both feature availability and selection change in that comparison. ANT's model-view values reconstructed from rounded figures remain approximate quantities in Supplement S2.

Recipient access changes the inference problem. PrivacyTrace's tool metadata/order yields attribute-inference ASR 14.53%, below its 17.97% schema-prior reference; adding arguments yields 27.71% [35]. MARK's combined F1 .971 also uses browser behavior, while its protocol/header features are available at the contacted website [6]. Encryption against transit observation does not hide an addressed request from its recipient.

These results identify where task information is observable. They lead to a testable question: with the initial instruction fixed and task-bearing feedback released later, how much recognition remains on the model link under a stated tool-side policy? Section 7 tests this question with constructed schemas and reports both the first-feedback and complete-execution views. Initial instructions, retained task fields and model responses are distinct possible contributors to the latter.

Defense consequence. Enumerate the services and representations available to the attacker, then evaluate the views remaining after an intervention. A joint observer can use a single-link decision rule, even when a particular fitted joint classifier scores lower. At the application boundary, minimize forwarded context while preserving the information needed for the task; at the transport boundary, evaluate visible sizes, timing and association. Their effects belong to separate contrasts.

5.2. Cover termination determines the execution constraint

Transport cover can terminate before business execution. Executed cover performs work and may affect resources, state or later decisions. The difference determines which utility must accompany a privacy measurement.

WRA's five research decoys reduce functional query recovery from .770 to .565 [2, Sec. 8.1]. Its utility scores 8.55 and 8.54 instead compare 100% and 10% destination visibility. Thus the utility observation does not describe the decoy intervention that produced the recovery reduction. WRA establishes additional research execution; it does not report harmful business-state writes. Matched protected-condition attacker examples also remain unresolved in the selected evidence.

The live OHTTP study refits a random forest on each protected capture and pairs six-class accuracy .30 with approximately thirteen-fold bytes and eleven-fold latency [3, Sec. 9.7]. Its separate nine-class .67 score belongs to another

TABLE 2. TRANSFER OF ESTABLISHED CONDITIONS TO AGENT EXECUTION. RELATIONSHIP LABELS CONCERN THE STATED MAPPING; EVIDENCE STRENGTH AND MISSING TESTS APPEAR SEPARATELY. A MISSING AGENT TEST DOES NOT REFUTE THE CLASSICAL RESULT.

Established relation and condition	Agent mapping / consequence	Relation	Agent evidence and remaining test
Unknown boundaries and background affect event recognition [45], [43], [44], [33].	Tool bursts, model waits and task intervals are different units. Evaluate proposed events with false alarms as well as classification.	Retained	ANT supplies stage boundaries; the local concurrent study tests particular detectors. Greater Agent localization difficulty is untested.
Defense evaluation must allow adaptation to the protected distribution [32], [16], [19].	Record protected training access and still-visible links; an unchanged learner family can refit.	Retained	Transformed and live OHTTP settings report adaptation in one preprint [3]; WRA's matched protected examples remain unresolved.
Coarse timing/volume can remain useful under specified padding [32]; serving streams expose length/gap signals [20], [12].	Name the shaping policy, receiving hop and Agent workload before carrying over a measured effect.	Untested generalization	Cloud and ANT measure model-view recognition [4], [9]. A general protected-Agent SSE claim is not tested by these observations.
An observation result is tied to the visible paths and representations [28], [34].	Feedback forwards tool information into later model requests across separately controlled services; evaluate every permitted residual view.	Extended	Published WRA tool-side evidence and preprint model/recipient results establish different exposed interfaces. Causal feedback attribution needs its own test.
STP tunnel endpoints remove cover before it reaches devices or servers [34, Sec. 7.1].	When cover executes as research, add original-task resource, state and feedback constraints to the traffic budget.	Extended	Published WRA reports executed research decoys and utility under a different intervention [2]. Our controlled LLM case separates task completion from state integrity (Section 7.1); it does not evaluate WRA writes.
Host traces can reconstruct distributed request paths [30], [31].	Instrumentation can provide ground truth; a transit attacker must infer from its own permitted observations.	Access mismatch	ANT stage labels and local task intervals are evaluator-provided. Host tracing alone cannot establish passive encrypted-stream localization.

comparison. This design point binds adapted inference to transport cost; matched original-task utility was not located.

Task completion is only one execution outcome. The requested refund can satisfy its target postcondition while another order, shared quota or ledger entry violates the application's state contract. Cover placement therefore determines what must be checked alongside completion. Section 7.1 illustrates this distinction with paired LLM executions and reports inference, state and resource outcomes together.

Defense consequence. An executed-cover design needs a semantic constraint alongside a traffic budget. It can terminate decoys before business execution, or isolate their resources, state and planner feedback. These are testable design choices: measure adapted inference, additional work, original-task completion and collateral state changes on the same protected runs. The published WRA text supports the intervention/utility distinction; the OHTTP refit/cost example depends on one preprint.

5.3. Execution changes delimit the two findings

A private recurring routine can be recognizable while its classifier transfers poorly to a new execution organization. In the communication-graph study, composition holdout keeps the nine workflow labels but changes their capability combinations: accuracy changes from about .67 to .18, with zero recall for seven classes [3, Sec. 9.12]. The cloud benchmark's graph/star change separately reports about .83 to .52 under its corresponding feature protocol [4]. Each axis depends on one preprint.

Implementation and failure handling can also change the observation. Trace reports macro-F1 .981 in its mixed-session model-attribution test and .815 on scaffold hold-out, retaining 1,010 of 1,050 planned clean runs [48]. AgentTrace's active probes reach 66.8% attribution from initial outcomes and 96.0% from ordered repairs at fifty probes [47]. Excluding failed sessions and observing recoverable errors are different operations. These source-specific conditions delimit what an attack or defense says about new workloads; they do not estimate a common filtering effect.

Finally, a traffic target retains its own security meaning. ANT labels benchmark risk, AgentTell tests action choice under a disclosure rule, and CHeaT measures prevention of a defined CTF objective [9], [8], [10]. Authorization additionally requires a principal and policy context. Two refunds that differ only in an unobserved approval can have identical traffic. This conditional argument identifies the extra policy input an authorization monitor needs.

TABLE 3. ATTACK FIELD MAP, PANEL A: TRANSIT OBSERVATIONS. SCORES RETAIN THEIR METRIC, TASK AND SUPPLIED UNIT; THEY ARE REPRESENTATIVE REPORTED OPERATING POINTS, NOT A CROSS-PAPER RANKING. THE MATRIX RECORDS ALL FORTY-FOUR SETTINGS. P: INSPECTED PREPRINT; V: INSPECTED PUBLISHER VERSION.

Interface / observer	Work / target	Observable features / learner	Evaluation unit	Reported result and boundary
Client–agent; transit	AgentPrint [P] [1]; hosted agent	Directional packet counts/bytes in 1,800 relative-time bins; CNN	Supplied browser interaction	Mixed-flow macro-F1: .8660 closed, .8477 open; 50 monitored agents. Behavior-F1 conflict is kept separate.
Agent–Web; transit	WRA [V] [2]; research query	Contacted domains and timing; ICL or fine-tuned LLM	Supplied research trace	Functional query recovery .770 with domains; .725 with domains plus timing in the compared visibility views. This is a reconstruction score.
Inter-agent / tools; transit	Communication graphs [P] [3]; workflow	Message/graph, size and timing descriptors; random forest	Workflow record	Nine-class accuracy .67; within-label composition holdout .18. The live defense test instead has six classes.
Agent–model; transit	Cloud benchmark [P] [4]; assigned task	247 traffic descriptors; random forest	Logged successful run	Ten-class balanced accuracy .853/.860 for graph/star coordination. Size-excluded feature protocols score .540/.587.
Agent–Web; transit	PersonaFingerprint [P] [5]; browsing persona	Signed lengths/interarrival times; CNN representation and prediction heads	Fixed packet window; session split	Per-site macro-F1 about .78–.91; pooled-summary discrepancy excluded from synthesis. Explicit other-class tests have a different population.
Combined egress, M or remaining links; transit	ANT [P] [9]; task, primitive, risk	Direction/length/time; 13 learners including DF and TikTok	Supplied episode or logged stage	Five-task accuracy .431–.727; 47-primitive accuracy .355–.462. Risk test: 20 positive/189 negative, requiring recall and false alarms.

6. Defense Mechanisms and Execution Costs

Defense placement determines both the signal changed and the party that must enforce the change.

6.1. What a defense changes

Table 6 organizes six intervention choices by enforcement point, trust, cost and residual signal. Association defenses hide a destination or its link to a user. Padding changes sizes, while scheduling changes temporal structure. Transport cover can terminate before business execution; execution cover adds real work. State isolation limits shared-resource signals, and content minimization limits what a recipient learns. These choices can be combined, but their costs and remaining views must be evaluated together.

6.2. Relating communication defenses to Agent execution

STP illustrates transport cover that can terminate at a tunnel endpoint [34]; Walkie-Talkie and FRONT provide shaping mechanisms [17], [18]. For the ticket Agent, this could mean padding the tool response at an egress proxy, pacing model tokens, or sending cover to a trusted terminator. Calling the refund tool as a decoy would instead cross the business-state boundary. The intervention must therefore specify both where cover is removed and what work it can trigger.

The measured design points occupy different parts of this space. WRA changes executed research, live OHTTP changes destination association, and the paired forwarding study changes the context sent to the model. Section 5 gives the source-specific recovery and cost results; Section 7 tests cover execution, forwarding and connection policies. Supplement S7 retains the complete defense map, including identity obfuscation and adversarial-Agent blocking as distinct objectives.

6.3. Choosing a layer in practice

Start from the private object and the observer's rights. Protect destination associations when contacted domains expose the query; protect visible model output sizes and timing when those carry the target; minimize sensitive arguments when the recipient is the observer. Then test what remains on other permitted links. A shared gateway changes which tasks contribute to a connection, so its connection policy belongs in the evaluation. For executed cover, bind privacy, token/tool use, requested-task completion and whole-state correctness to the same protected runs. These choices turn the risk map into a deployment-specific protection plan.

7. Controlled Examples of the Execution Boundaries

The cover study illustrates joint measurement of inference, target-task outcomes, collateral state and cost. Its adverse control exercises the difference between the two outcome predicates. The separate forwarding pilot examines an unresolved information path; the studies retain their own populations and inferential results.

TABLE 4. ATTACK FIELD MAP, PANEL B: CONTACTED RECIPIENTS AND ACTIVE AUDITORS. THESE OBSERVERS SEE REQUESTS ADDRESSED TO THEM OR ISSUE PROBES; THEIR RESULTS DO NOT MEASURE ENCRYPTED-TRANSIT INFERENCE. E: INSPECTED EXTENDED PREPRINT WITH A SEPARATELY VERIFIED PUBLISHED SHORT VERSION. NR: NO RECOGNITION RATE REPORTED IN THE MAPPED SETTING. ASR: ATTACK SUCCESS RATE.

Interface / access	Work / target	Observable features / learner	Reported result and evaluation boundary
Web recipient	MARK [P] [6]; client identity	TLS/HTTP2 and headers; decision tree	F1 .725 for TLS/HTTP2, .799 for headers; .971 combined with browser instrumentation, a richer view. First-pass trials.
Web recipient	Fayolle et al. [P] [7]; client / access	Protocol signatures; richer behavior classifier and site challenges	Fingerprint analysis excludes failed visits. Access success is separately verified through form/cookie/page identifiers; not a privacy score.
Documentation recipient	HTTP signatures [E] [49]; service/client	Request method/version, headers and origin; descriptive comparison	Nine coding agents and six assistants, three trials each; NR. Request signatures are observations, not a held-out identification rate.
Terminal recipient	Trace [P] [48]; model family	Command TF-IDF unigrams/bigrams; linear SVM	Seven families; 2,028 clean/injected sessions: macro-F1 .981; held-out scaffold .815. A separate proprietary-scaffold test reports 78% accuracy.
Tool interface; active audit	AgentTrace [P] [47]; model identity	Serialized arguments, parser errors and repair order; Markov likelihood	Six models, 50 held-out probes per trial: 96.0% accuracy; 10 probes: 92.8%. Repeated stochastic trials reuse probe identities.
Web recipient; active choice	AgentTell [P] [8]; prior-origin fact	Which offered action is requested; loaded/cold comparison	Matching choice in 61.1% of 7,630 loaded sessions; the leakage measure subtracts cold matching. Actions share a completable probe task.
Tool recipient / gateway	PrivacyTrace [P] [35]; profile attribute	Tool metadata/order; content-added LLM inference	Metadata ASR 14.53%, schema prior 17.97%; arguments 27.71%. 68,000 attribute slots are clustered within 4,000 trajectories.
Contacted asset; active challenge	CHeaT [V] [10]; agent presence	Planted URL following; separate password-output control	Presence-detection mechanisms, NR for natural-traffic classification. Capture-the-flag prevention is a distinct endpoint (Supplement S7).
SSH recipient; active probes	LLM Agent Honeypot [P] [51]; possible agent	Prompt compliance and response-time heuristic	Three-month deployment: eight candidate agents among 8,130,731 interactions. No externally verified field labels or detection precision/recall.

7.1. Joint traffic and execution measurements

The study compares six cover policies on matched synthetic support tickets. Two fixed-weight models, Qwen3:8B and Llama3.1:8B, each run 300 fixtures per condition: 120 training, 60 validation and 120 test, balanced across six actions. The 3,600 episodes have terminal results. Fixtures and cover draws are paired across conditions; request phrasings recur across splits.

Execution and observation. The models choose checks, business actions, objects, recovery and termination through LangGraph [56]. A controller fetches the ticket and enforces prerequisites without supplying the correct action. Seven local MCP Streamable HTTP destinations use TLS 1.3/HTTP/2, a common `/mcp` path and request/response padding quanta of 4,096/16,384 bytes. The observer receives destinations and bidirectional ciphertext byte counts within supplied episode and call windows. Timing and model traffic are outside this view.

A frozen schedule inserts one or two random decoys among early tool calls. The conditions are no cover, proxy discard, read cover, isolated execution, live write, and isolated execution with visible responses. Proxy discard computes on a disposable copy; isolated execution retains a separate clone. Live writes deliberately modify the primary fixture, including other orders or shared quota. The visible control sends clearly marked cover responses from the clone to the planner. Other cover responses are hidden. The live-write arm is an adverse control selected to exercise collateral changes. Proxy discard also computes a disposable response; it is not a pure transport-only dummy.

Matched endpoints and attacks. *Goal completion* checks the requested action/object, prerequisites and target postcondition. *State integrity* checks the entire business state against the single intended-action reference. We also count completed tasks with state errors directly; the two predicates are not nested. Recognition predicts the requested action including failed executions. Logistic regression (LR) and random forest (RF) train within each condition, with validation selecting the primary attacker. Frozen and destination-position attacks are retained alongside them.

Each endpoint has a prespecified twelve-test Holm family: five cover-minus-baseline contrasts and visible-minus-hidden for each model. Conditional intervals resample paired fixtures by action, holding fitted attacks fixed. Supplement S8 contains every attack and contrast, state replay and a worked evaluation protocol.

Completion and collateral effects diverge. Under live writes, 95 of 102 completed Qwen tasks and 73 of 76

TABLE 5. CONDITIONAL RISK MAP. A ROW APPLIES WHEN ITS ACCESS AND EXECUTION CONDITIONS HOLD. THE FINAL COLUMN STATES A PROTECTION PRIORITY OR DESIGN HYPOTHESIS, NOT A DEMONSTRATED MITIGATION. SCORES, TARGETS AND RIGHTS DIFFER ACROSS ROWS, SO THERE IS NO GLOBAL DANGER RANKING.

Interface and signal	Condition and evidenced inference	Protection consequence
Tool egress: selected domains and timing	The observer can associate contacted names with a supplied research trace. WRA recovers query meaning; linked histories use a separate persona endpoint [2].	Prioritize destination association when the private target is research intent. Reduced name visibility needs a residual traffic assessment; executed decoys need task-utility constraints.
Model link: lengths, directions and timing	Model traffic is visible and associated with the evaluated task. ANT and the cloud benchmark measure task information in this view [9], [4]. Initial instructions and later feedback coexist.	Tool-only protection leaves this view to be assessed. Compare model-only and joint adaptive attacks; isolate feedback before choosing a feedback-specific defense.
Coordination: message graph, volume and sequence	Workflow records or coordination traffic expose execution organization. Composition and topology transfer tests change that organization under known task labels [3], [4].	Target observable scheduling and association, not just one payload size. Evaluate changed compositions and topology separately from recurrent-workflow privacy.
Client link: packet counts, directions and bursts	AgentPrint's supplied browser interactions support hosted-agent identity inference [1]. The result presumes those observation units and identities.	Client-facing traffic needs its own protection assessment even if tool links are hidden. Task discovery and new identities require separate tests.
Contacted recipient: headers, commands or arguments	MARK, Trace and PrivacyTrace use recipient-visible representations. In PrivacyTrace, metadata alone is below its prior, while adding arguments improves attribute inference [6], [48], [35].	Minimize sensitive request content and assess implementation signatures. Transport encryption does not conceal the recipient's intended plaintext.
Active probing: challenges, choices and repairs	A party can elicit responses and observe later behavior. AgentTrace attributes models from repairs; AgentTell observes a choice under a disclosure rule [47], [8].	Test error normalization and restrictions on cross-context disclosures while retaining task recovery. These hypotheses need active-access evaluations.

completed Llama tasks have collateral state errors. The corresponding no-cover counts are zero of 114 and thirteen of 82. Whole-state integrity falls from 116/120 to 7/120 and from 69/120 to 3/120. Both paired reductions survive correction; completion falls by 10.0 and 5.0 percentage points, with only Qwen's contrast significant. The adverse writes strongly predispose the direction of state loss. The counts quantify disagreement between the target-task predicate and the broader fixture contract. They neither discover a weakness in state-aware benchmarks nor estimate ordinary cover harms.

Residual recognition and execution cost. For Qwen, hidden isolated execution completes the same 114/120 tasks as no cover, with state integrity 115/120 versus 116/120 and 1.34 times the tool-call bytes. The highest observed score across the tested attacks falls from 118/120 to 77/120. That latter score comes from the last-destination rule, above the selected learner's 61/120. It exposes the schedule's unprotected tail and gives a more demanding account of residual recognition. At this budget, Qwen's selected attack scores are 62, 62, 61 and 62 for discarded, read, isolated and live-write cover. Live writes show no additional recognition reduction in these point estimates. These observations do not establish equivalence among policies. Across both models, all ten primary cover-versus-baseline recognition contrasts survive correction; substantial recognition remains.

Visible and hidden isolated responses are identical within test pairs. Visibility increases mean prompt tokens by 18.1% and 18.9%, while all primary recognition, goal and state contrasts have adjusted $p = 1$. The experiment leaves their average effect unresolved. Paired fixtures remain matched despite planner path variation; without repeated runs of each condition, that variation cannot be separated into a stable visibility effect and execution noise. Supplement S8 reports the paths.

7.2. Unresolved feedback effects and event detection

The separate pilot holds initial messages fixed and varies whether supplementary tool records enter the next model request. *Full* forwards the whole response; *Projected* retains task-required fields. The synthetic records deliberately carry action information. Two fixed 8B models perform 3,120 local LangGraph/MCP executions over streaming TLS 1.3/HTTP/2; each test condition contains 120 tasks in twenty nonempty and four empty sessions.

First-feedback LR recognition changes from 43 to 23 correct for Qwen and from 40 to 22 for Llama. All eight primary local comparisons, including post-feedback versus

TABLE 6. DEFENSE DESIGN SPACE: ENFORCEMENT AUTHORITY, COSTS TO MEASURE AND REMAINING EXPOSURE. SOURCE-SPECIFIC MEASUREMENTS APPEAR IN THE TEXT AND SUPPLEMENT S7.

Layer / mechanism	What must be trusted or controlled	Costs to measure together	Residual exposure and practical decision
Naming / association: encrypted name resolution, relays, OHTTP	Name resolver or relay placement; the parties able to link the user to the destination	Added hops, bytes, latency and availability	Shape can survive name hiding. Choose this layer when destination association is the private target; then test remaining observers.
Representation: message, TLS-record or packet padding	Component that controls the final visible size; padding survives later encoding	Added bytes, record/packet count, fragmentation	Timing, count and other links remain. State the padding layer and test downstream captures.
Scheduling: batching, pacing, fixed-rate cover	Sender/relay controls output schedule and cover lifetime	First-response delay, completion delay, bandwidth during idle periods	Admission, termination and overflow can reveal activity. Budget these events as part of the visible trace.
Transport cover: dummy exchanges discarded at a tunnel endpoint	Trusted cover terminator; no dispatch into business tools	Cover bandwidth, queueing, resource use	Other endpoints still see their inputs. Prefer this boundary when the aim is traffic shaping without extra business work.
Execution cover: extra research or tool calls	Tool privileges, isolated state and whether decoy feedback reaches the planner	Model tokens, tool quota, task success, state changes and latency	A decoy can affect the real execution. Measure privacy and original-task utility on the same protected runs.
Shared-state isolation / minimization	Tenant separation, cache keys and information passed between services	Lost cache reuse, compute, latency, task accuracy	Isolation removes a sharing channel, not every traffic signal. Minimize tool arguments and feedback according to each recipient's role.

TABLE 7. EXECUTED COVER: ALL HELD-OUT CONDITIONS, 120 PAIRED FIXTURES PER MODEL AND CONDITION. RECOGNITION COLUMNS COUNT CORRECT REQUESTED-ACTION PREDICTIONS; CHANCE IS 20/120. THE LEARNER IS SELECTED USING VALIDATION LR/RF ACCURACY. THE POSITION RULE REMAINS VISIBLE WHEN IT IS STRONGER. GOAL, WHOLE-STATE INTEGRITY AND COMPLETED TASKS WITH STATE ERRORS ARE SEPARATE ENDPOINTS.

Model	Cover policy	Selected	Last dest.	Goal	State	Goal + error	Call bytes ×
Qwen3:8B	No cover	118	116	114	116	0	1.00
Qwen3:8B	Proxy discard	62	77	114	116	0	1.34
Qwen3:8B	Read cover	62	77	114	115	0	1.34
Qwen3:8B	Isolated, hidden	61	77	114	115	0	1.34
Qwen3:8B	Live write	62	75	102	7	95	1.34
Qwen3:8B	Isolated, visible	63	77	113	114	1	1.35
Llama3.1:8B	No cover	113	79	82	69	13	1.00
Llama3.1:8B	Proxy discard	79	67	80	68	13	1.29
Llama3.1:8B	Read cover	72	67	84	71	14	1.28
Llama3.1:8B	Isolated, hidden	82	67	83	70	13	1.28
Llama3.1:8B	Live write	80	65	76	3	73	1.29
Llama3.1:8B	Isolated, visible	81	68	83	73	11	1.29

prefix, exceed .05 after Holm correction (minimum .0508). Complete-execution LR changes from 46 to 44 and from 62 to 33. Projection changes content, length and execution order together, so the pilot leaves the contribution of feedback content and end-to-end protection unresolved. The hosted-alias retest also has no corrected primary rejection. Supplement S1 retains every attack, phase, interval and execution outcome.

Concurrent runs additionally test continuous detectors with visible connection identities and no task windows. Their Shared-condition precision is 4.3–18.6%, with 8.21–44.79 false events per test session at IoU .5. A constant all-active baseline ranks above learned detectors at IoU .3 and below them at .5. This diagnoses detector and matching-rule behavior in the constructed workload. Full .3/.5/.7 results, background definitions and mixed shaping outcomes remain in Supplement S1. These diagnostics guide evaluation design; they do not measure general Agent localization difficulty.

8. Evaluating Both Boundaries

An intervention needs one record linking the attacker's remaining observations to the original task and allowed

collateral effects. Table 8 shows why this joint record matters: among four query/workflow privacy settings from two sources, adaptation is documented in two and matched original-task utility in one, with no setting documenting both. The adapted settings share a preprint; the utility observation is WRA's published visibility comparison. These are descriptive counts of inspected configurations.

8.1. A minimal evaluation procedure

The following procedure connects established evaluation guidance [52], [15], [16] to the information and action boundaries. It is a proposed reporting procedure, illustrated by the controlled cover case. The six observation-contract fields in Section 3 specify an inference setting; the eight checklist items organize the broader study report, including its outcomes, uncertainty and provenance.

1) **Declare the target and permitted observations.** Name the private fact, visible model/tool/recipient interfaces, access rights and any supplied execution windows. Track where tool results are forwarded.
2) **Declare the allowed execution effects.** Specify the original goal, other objects that may change, resource budgets and what cover may return to the planner. Use business invariants that permit legitimate concurrent updates; exact state equality fits only a correspondingly constrained fixture.
3) **Pair protection, attacks and outcomes.** Evaluate the baseline and intervention on matched tasks. Train permitted attackers on protected executions and include attacks that exploit the cover schedule. Record goal completion, completed tasks violating the state contract, and bytes, delay and resource use on those same runs.
4) **Make the decision against stated tolerances.** Report attack-specific inference and uncertainty together with task, state and resource tolerances fixed before evaluation. A completed task with a state change outside the declared contract fails its utility requirement. Successful inference on another permitted view defeats a claim of containment across those views. An unsuccessful tested attack supplies no general secrecy guarantee.

Worked application. The cover study binds requested-action recognition to a tool-metadata observer and checks both target completion and the whole synthetic fixture. Its adverse live-write arm separates target completion from collateral effects, while the other policies record residual recognition and cost under the same observer. Table 7 keeps the stronger position rule visible. Supplement S8 derives the fixture contract and fills each procedure step. The application demonstrates joint reporting; model/timing views and deployment acceptance thresholds require additional evidence.

8.2. Corpus availability and reusable forms

Three identity-obfuscation settings, two adversarial-Agent blocking settings and thirty-five without a protection comparison have different eligibility from the four privacy settings. The access map separately assigns twenty-two transit, ten recipient, eleven active and one downstream preference setting. Supplement S4.1 reports outcome/failure encoding sensitivity with each encoding's own denominator. Supplement S5 provides the eight-field checklist, its correspondence to established guidance and filled WRA, OHTTP and ANT examples. Each form records evidence, a missing item or inapplicability.

9. Synthesis and Research Agenda

9.1. Which conclusions depend on which sources

The evidence supports different parts of the two-boundary account at different strengths. Published WRA links tool destinations to query recovery and separates executed-decoy recovery from the visibility condition's utility. Model-view measurements, live OHTTP refitting and the individual composition, topology and scaffold transfer axes depend on preprints. Multiple settings from one source retain that shared dependence.

Source withdrawal follows thirty-three statements across seven comparisons and twenty removal scenarios. In the strict published-text subset, WRA and CHeaT retain query recovery, the decoy/visibility distinction and a defined prevention outcome. Removing a transfer source removes evidence for that axis. The supplement gives the full accounting. The local cover study illustrates outcome coverage under an adverse control; the forwarding pilot supplies an unresolved causal comparison. Neither substitutes for missing source coverage.

9.2. Tests that would change a protection decision

Table 9 turns the synthesis into four discriminating tests: transfer across execution changes, separation of forwarding effects, cover under business constraints, and inference after event discovery. For the ticket application, useful progress means meeting both boundaries: limiting the observations carrying task information while keeping decoys within specified state and resource budgets. A production claim requires tolerances for legitimate updates and a workload that exercises them.

10. Validity, Ethics and Availability

Evidence scope. The corpus combines query retrieval, selected seeds and citation tracing. Index failures and missed query terms limit coverage. Extraction is author-performed with AI assistance. Two dependent, non-blind AI recodings diagnose the manual; they provide no independent

TABLE 8. EVIDENCE LOCATED IN THE RETAINED RECORDS. COUNTS USE ELIGIBLE SETTINGS; THE SECOND COLUMN ALSO GIVES DISTINCT SOURCES. "NOT LOCATED" LEAVES EVIDENCE UNRESOLVED WITHIN THE INSPECTED SCOPE. THESE COUNTS DESCRIBE AVAILABILITY, NOT STUDY QUALITY OR CHECKLIST VALIDITY.

Question	Eligible settings / sources	Located	Supporting settings and consequence
Protected-condition attack adaptation	4 / 2	2/4	Communication graphs: transformed captures and live OHTTP. Both come from one preprint.
Original-task utility under the same intervention	4 / 2	1/4	WRA visibility comparison. It does not price WRA's executed-decoy condition.
Adaptation and original-task utility in one setting	4 / 2	0/4	Adaptation and utility are located in different records; transport cost alone does not fill the task-outcome field.
Original-task utility for executed cover	1 / 1	0/1	WRA's five research decoys report recovery and extra research; matched original-task utility is not located.

TABLE 9. TESTABLE OPEN QUESTIONS DERIVED FROM THE SYNTHESIS. Q1 SPECIFIES TRANSFER TESTS. Q2 SEPARATES THE FORWARDING MEDIATORS; Q3 EXTENDS THE CONTROLLED COVER STUDY TO APPLICATION CONSTRAINTS; Q4 EXTENDS LIVE DETECTION TO NEW WORKLOADS AND DEPLOYMENTS. EACH GIVES A RESULT THAT COULD SUPPORT OR REVISE THE PROPOSED EXPLANATION.

Question and unresolved mechanism	Controlled comparison	Discriminating result
Q1. Which signals survive execution changes? Do attacks identify recurring task structure or features tied to its implementation?	Keep task labels and observation rights fixed. Hold out compositions, coordination and client/scaffold versions separately, then jointly. Compare destination, volume/timing, sequence and recipient-visible feature families under declared frozen and adapted training.	Identify a feature family whose inference survives specified changes, or a failure tied to one changed condition. Report per-axis and joint transfer with attempted-run coverage; a pooled average cannot establish invariance.
Q2. How does feedback reach model traffic? Which part is content, length or later action structure?	Fix initial instructions and task-required fields. Pair length-matched neutral padding with validity-preserving replacement of supplementary content; hold encoding and follow-up policy fixed where possible. Vary record count separately. Compare prefix, first-feedback and complete model views under tool-only and joint protection.	Additional held-out inference after feedback, with controls that exclude the competing path, supports that attribution. Report an effect interval against a prespecified meaningful-effect threshold. A nonsignificant difference alone does not rule out leakage.
Q3. Can cover preserve task semantics? When should it stop at transport or execute in isolation?	Extend the six-condition LLM case to production tool contracts and legitimate concurrent updates. Compare no cover, transport termination and isolated execution under declared resource budgets. Refit permitted attackers; measure bytes, delay, original-task completion, quota use, state changes and feedback entering the planner.	A privacy–utility operating point satisfying prespecified state and task tolerances supports the design. Resource exhaustion, changed business state or impaired completion identifies the constraint it violates.
Q4. Which inference survives task discovery? Does cross-link association recover information hidden by waits and multiplexing?	Train on one concurrency/load and test new compositions, loads and gateways. Compare tool-only and associated tool/model views with observable flow identifiers; keep ground-truth task and stream mappings outside attacker inputs. Include real unrelated traffic and constant-output detector references.	Measure interval recall, false alarms per unit time and correctly localized inference together. Gains must survive changed workloads and loads at a stated false-alarm budget.

human reliability. The outcome disagreements led to source-specific corrections, with original codes and sensitivity retained. Source locators and withdrawal analysis make these dependencies inspectable.

The executed-cover case uses synthetic actions, local services, reused phrasings and supplied episode/call windows. Its intervals condition on fitted attacks and the chosen fixtures. Adverse writes are selected to exercise collateral effects, so the measured losses do not estimate production prevalence or WRA's research-decoy harms. Llama's baseline already has goal and state errors. Clearly labelled visible cover responses leave average endpoint effects unresolved. Business-state invariants, stronger attacks and other observation views require their own evidence.

The forwarding pilot uses deliberately category-bearing records and twenty nonempty test-session clusters. Its corrected primary comparisons remain inconclusive; content, length and execution order change together. Session resampling and LR refitting address variation within the selected workload and learner family. Live background traffic is synthetic. The hosted retest changes backend and network together and uses an unpinned alias. Supplement S1 reports

these populations separately; earlier experiments and unknown responses remain in the artifact archive.

Ethics and assistance. Inputs are synthetic, business actions update isolated fixtures, and capture filters restrict collection to the experiment. Remote model use has the account holder's authorization; provider-specific research permission and applicable terms remain unverified. Known failures and unknown responses are retained separately. No human-participant experiment, customer operation or hosted-provider vulnerability finding enters the study. AI tools assisted source extraction, implementation and writing; saved observations and executable analyses support the measurements.

Traffic inference can enable unwanted profiling as well as monitoring. The review package supplies extraction records, synthetic metadata, features, predictions and offline analyses. Synthetic planner records and fixtures permit state replay. Credentials, private transcripts, raw captures and third-party full texts are excluded; the remaining methods still carry misuse potential. A protection summary should present adapted attacks, permitted views and original-task outcomes together, since a frozen attack's decline can overstate protection.

Availability. The accompanying package connects the forty-four-setting matrix, transfer relations, eligibility ledger and source locators to the claims. It includes the prior evidence archive, synthetic predictions and outcome records, uncertainty scripts and manifests. Excluded captures are needed for raw-to-metadata reconstruction, and a mutable remote alias cannot supply fixed-weight reproducibility. Compilable LaTeX and both PDFs accompany these materials.

11. Conclusion

Agent communication defenses operate across service and execution boundaries. The synthesis links each signal to an observer with specific access, and each cover policy to the work and state it can affect. Its concrete consequences are to evaluate every permitted residual view and to bind privacy, task outcomes and resource cost to the same intervention. The controlled study illustrates this joint record; the open agenda asks when feedback adds recoverable information and when practical cover satisfies a business contract. These questions preserve the Agent execution focus while making the supporting evidence and remaining tests explicit.

References

[1] Y. Zhang, X. Deng, Z. Gu, Y. Chen, K. Xu, Q. Li, and J. Wu, "Exposing LLM user privacy via traffic fingerprint analysis: A study of privacy risks in LLM agent interactions," *arXiv preprint arXiv:2510.07176*, 2025. [Online]. Available: https://arxiv.org/abs/2510.07176v1

[2] H. Jeong, M. Teymoorianfard, A. Kumar, A. Houmansadr, and E. Bagdasarian, "Network-level prompt and trait leakage in local research agents," in *35th USENIX Security Symposium*. USENIX Association, 2026, pp. 6485–6504. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/jeong

[3] B. Dangol, "From privacy to workflow integrity: Communication-graph metadata in autonomous agent interoperability," *arXiv preprint arXiv:2606.07150v3*, 2026. [Online]. Available: https://arxiv.org/abs/2606.07150v3

[4] S. Pouryousef, J. Lopez, S. R. Nowmi, M. M. Kamol, M. Hossain, M. Tran, and M. S. Rahman, "A large-scale benchmark and risk assessment of traffic analysis attacks on cloud LLM services," *arXiv preprint arXiv:2609.31877v2*, 2026. [Online]. Available: https://arxiv.org/abs/2609.31877v2

[5] C. Song, H. Wang, and R. Martin, "PersonaFingerprint: Measuring persona inference on modern websites with LLM-driven browsing," *arXiv preprint arXiv:2605.15962v1*, 2026. [Online]. Available: https://arxiv.org/abs/2605.15962v1

[6] D. Kang, H. Jeong, J. Sheffey, P. Datta, and A. Houmansadr, "Whose agent are you? multi-layer fingerprinting and attribution of autonomous web agents," *arXiv preprint arXiv:2606.20910v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.20910v1

[7] I. Fayolle, S. Bouhenniche, S. Pélissier, P. Laperdrix, C. Maurice, and W. Rudametkin, "On the internet, nobody knows you're an LLM bot: Unmasking web agents with multi-layer fingerprinting," *arXiv preprint arXiv:2606.30119v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.30119v1

[8] A. Shahriar, M. N. Rahman, S. Ahmed, F. Sadeque, and M. R. Parvez, "AgentTell: Behavioural side-channel leakage in browser-use agents," *arXiv preprint arXiv:2609.32915v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.32915v1

[9] F. Li, X. Gao, Z. Liu, T. Li, C. Fu, Z. Wang, and K. Xu, "ANT: A multi-granularity network traffic dataset and benchmark for agents behavior auditing," *arXiv preprint arXiv:2610.06514v1*, 2026. [Online]. Available: https://arxiv.org/abs/2610.06514v1

[10] D. Ayzenshteyn, R. Weiss, and Y. Mirsky, "Cloak, honey, trap: Proactive defenses against LLM agents," in *34th USENIX Security Symposium*. USENIX Association, 2025, pp. 8095–8114. [Online]. Available: https://www.usenix.org/conference/usenixsecurity25/presentation/ayzenshteyn

[11] L. Song, Z. Pang, W. Wang, Z. Wang, X. Wang, H. Chen, W. Song, Y. Jin, D. Meng, and R. Hou, "The early bird catches the leak: Unveiling timing side channels in LLM serving systems," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 11 431–11 446, 2025, cited version: arXiv:2409.20002v5. [Online]. Available: https://ieeexplore.ieee.org/document/11206380/

[12] G. McDonald and J. B. Or, "Whisper leak: a side-channel attack on large language models," *arXiv preprint arXiv:2511.03675*, 2025, cited version: arXiv:2511.03675v1. [Online]. Available: https://arxiv.org/abs/2511.03675v1

[13] T. Zhang, G. Saileshwar, and D. Lie, "Time will tell: Timing side channels via output token count in large language models," *arXiv preprint arXiv:2412.15431*, 2024, cited version: arXiv:2412.15431v1. [Online]. Available: https://arxiv.org/abs/2412.15431v1

[14] G. Wu, Z. Li, Y. Zhang, Z. Zhang, J. Niu, Y. Wu, and Y. Zhang, "Cacheprune: Privacy-aware and fine-grained KV cache sharing for efficient LLM inference," *arXiv preprint arXiv:2605.23640v1*, 2026. [Online]. Available: https://arxiv.org/html/2605.23640v1

[15] M. Juarez, S. Afroz, G. Acar, C. Diaz, and R. Greenstadt, "A critical evaluation of website fingerprinting attacks," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2014, pp. 263–274.

[16] N. Mathews, J. K. Holland, S. E. Oh, M. S. Rahman, N. Hopper, and M. Wright, "SoK: A critical evaluation of efficient website fingerprinting defenses," in *IEEE Symposium on Security and Privacy*, 2023.

[17] T. Wang and I. Goldberg, "Walkie-talkie: An efficient defense against passive website fingerprinting attacks," in *26th USENIX Security Symposium*, 2017. [Online]. Available: https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/wang-tao

[18] J. Gong and T. Wang, "Zero-delay lightweight defenses against website fingerprinting," in *29th USENIX Security Symposium*. USENIX Association, 2020, pp. 717–734. [Online]. Available: https://www.usenix.org/conference/usenixsecurity20/presentation/gong

[19] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Advances in Neural Information Processing Systems*, vol. 33, 2020. [Online]. Available: https://arxiv.org/abs/2002.08347v2

[20] R. Weiss, D. Ayzenshteyn, G. Amit, and Y. Mirsky, "What was your prompt? a remote keylogging attack on ai assistants," in *33rd USENIX Security Symposium*, 2024. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/weiss

[21] N. Carlini and M. Nasr, "Remote timing attacks on efficient language model inference," *arXiv preprint arXiv:2410.17175v1*, 2024. [Online]. Available: https://arxiv.org/html/2410.17175v1

[22] D. Ahmed, A. Sabir, and A. Das, "Spying through your voice assistants: Realistic voice command fingerprinting," in *32nd USENIX Security Symposium*, 2023, pp. 2419–2436. [Online]. Available: https://www.usenix.org/conference/usenixsecurity23/presentation/ahmed-dilawer

[23] Model Context Protocol, "Transports," 2025, protocol version 2025-06-18. [Online]. Available: https://modelcontextprotocol.io/specification/2025-06-18/basic/transports

[24] M. Thomson and C. Benfield, "HTTP/2," RFC 9113, 2022. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9113.html

[25] E. Rescorla, "The transport layer security (tls) protocol version 1.3," RFC 8446, 2018. [Online]. Available: https://www.rfc-editor.org/rfc/rfc8446.html

[26] W. Lugoloobi, S. Marro, J. Magomere, J. Wright, and C. Russell, "Known by their actions: Fingerprinting LLM browser agents via UI traces," *arXiv preprint arXiv:2605.14786v1*, 2026. [Online]. Available: https://arxiv.org/abs/2605.14786v1

[27] J. Roh, E. Bagdasarian, H. Haddadi, and A. S. Shamsabadi, "SPILLage: Agentic oversharing on the web," *arXiv preprint arXiv:2602.13516v1*, 2026. [Online]. Available: https://arxiv.org/abs/2602.13516v1

[28] S. Chen, R. Wang, X. Wang, and K. Zhang, "Side-channel leaks in web applications: A reality today, a challenge tomorrow," in *IEEE Symposium on Security and Privacy*, 2010. [Online]. Available: https://staff.ie.cuhk.edu.hk/~khzhang/my-papers/2010-oakland-side-channel.pdf

[29] P. Chapman and D. Evans, "Automated black-box detection of side-channel vulnerabilities in web applications," in *ACM Conference on Computer and Communications Security*, 2011. [Online]. Available: https://www.cs.virginia.edu/~evans/pubs/ccs2011/

[30] P. Reynolds, J. L. Wiener, J. C. Mogul, M. K. Aguilera, and A. Vahdat, "WAP5: Black-box performance debugging for wide-area systems," in *International World Wide Web Conference*, 2006. [Online]. Available: https://cseweb.ucsd.edu/~vahdat/papers/wap5-www06.pdf

[31] B. C. Tak, C. Tang, C. Zhang, S. Govindan, B. Urgaonkar, and R. N. Chang, "vPath: Precise discovery of request processing paths from black-box observations of thread and network activities," in *USENIX Annual Technical Conference*, 2009. [Online]. Available: https://www.usenix.org/conference/usenix-09/vpath-precise-discovery-request-processing-paths-black-box-observations-thread

[32] K. P. Dyer, S. E. Coull, T. Ristenpart, and T. Shrimpton, "Peek-a-boo, i still see you: Why efficient traffic analysis countermeasures fail," in *IEEE Symposium on Security and Privacy*, 2012. [Online]. Available: https://pages.cs.wisc.edu/~rist/papers/trafanal.pdf

[33] R. Trimananda, J. Varmarken, A. Markopoulou, and B. Demsky, "Packet-level signatures for smart home devices," in *Network and Distributed System Security Symposium*, 2020. [Online]. Available: https://www.ndss-symposium.org/ndss-paper/packet-level-signatures-for-smart-home-devices/

[34] N. Apthorpe, D. Y. Huang, D. Reisman, A. Narayanan, and N. Feamster, "Keeping the smart home private with smart(er) IoT traffic shaping," *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 3, pp. 128–148, 2019. [Online]. Available: https://petsymposium.org/popets/2019/popets-2019-0040.pdf

[35] D. Duan, Q. Liu, Y. Liu, N. Zou, X. Hu, and D. Liu, "How much can tool calls reveal about you? privacy leakage from tool-call traces," *arXiv preprint arXiv:2610.41885v1*, 2026. [Online]. Available: https://arxiv.org/abs/2610.41885v1

[36] Y. Yao, Z. Wang, H. Cheng, Y. Cheng, H. Du, and X.-Y. Li, "IntentMiner: Intent inversion attack via tool call analysis in the model context protocol," *arXiv preprint arXiv:2512.14166v2*, 2026. [Online]. Available: https://arxiv.org/abs/2512.14166v2

[37] P. Sirinam, M. Imani, M. Juarez, and M. Wright, "Deep fingerprinting: Undermining website fingerprinting defenses with deep learning," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2018. [Online]. Available: https://arxiv.org/abs/1801.02265v5

[38] S. Bhat, D. Lu, A. Kwon, and S. Devadas, "Var-CNN: A data-efficient website fingerprinting attack based on deep learning," *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 4, pp. 292–310, 2019. [Online]. Available: https://petsymposium.org/popets/2019/popets-2019-0070.php

[39] M. Bishop, "HTTP/3," RFC 9114, 2022. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9114.html

[40] J. Iyengar and M. Thomson, "QUIC: A UDP-based multiplexed and secure transport," RFC 9000, 2021. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9000.html

[41] C. Song, D. D. M. Mekala, H. Wang, and R. P. Martin, "Redefining website fingerprinting attacks with multi-agent LLMs," *Proceedings on Privacy Enhancing Technologies*, vol. 2026, no. 3, pp. 688–702, 2026. [Online]. Available: https://www.petsymposium.org/popets/2026/popets-2026-0101.php

[42] S. Zavrak, "Content-aware attack detection in LLM agent tool-call traffic: An empirical study of features, architectures, and evaluation protocols," *arXiv preprint arXiv:2605.11053v3*, 2026. [Online]. Available: https://arxiv.org/abs/2605.11053v3

[43] Z. Guan, G. Xiong, G. Gou, Z. Li, M. Cui, and C. Liu, "BAPM: Block attention profiling model for multi-tab website fingerprinting attacks on Tor," in *Annual Computer Security Applications Conference*, 2021.

[44] X. Deng, Q. Yin, Z. Liu, X. Zhao, Q. Li, M. Xu, K. Xu, and J. Wu, "Robust multi-tab website fingerprinting attacks in the wild," in *IEEE Symposium on Security and Privacy*, 2023, pp. 1005–1022.

[45] T. Wang and I. Goldberg, "On realistically attacking Tor with website fingerprinting," *Proceedings on Privacy Enhancing Technologies*, vol. 2016, no. 4, pp. 21–36, 2016.

[46] J. Hayes and G. Danezis, "k-fingerprinting: A robust scalable website fingerprinting technique," in *25th USENIX Security Symposium*, 2016, pp. 1187–1203. [Online]. Available: https://www.usenix.org/conference/usenixsecurity16/technical-sessions/presentation/hayes

[47] G. Zhao, H. Wu, and X. Zhang, "AgentTrace: Execution-level fingerprinting of black-box LLM agents," 2026, version 1. [Online]. Available: https://arxiv.org/abs/2610.32092v1

[48] M. Ediga and S. Chattopadhyay, "Trace: Unmasking AI attack agents through terminal behavior fingerprinting," 2026, version 1. [Online]. Available: https://arxiv.org/abs/2605.01186v1

[49] O. Borysenko, "Developer experience with AI coding agents: HTTP behavioral signatures in documentation portals," 2026, extended version 3. [Online]. Available: https://arxiv.org/abs/2604.02544v3

[50] ——, "Documentation retrieval by AI coding agents: An empirical study of HTTP behavioral signatures," in *Proceedings of the 1st International Workshop on Agentic AI for Next-Generation Software Development*. ACM, 2026, pp. 168–169.

[51] Reworr and D. Volkov, "LLM agent honeypot: Monitoring AI hacking agents in the wild," 2025, version 2; first submitted 2024. [Online]. Available: https://arxiv.org/abs/2410.13919v2

[52] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in *31st USENIX Security Symposium (USENIX Security 22)*. USENIX Association, 2022, pp. 3971–3988. [Online]. Available: https://www.usenix.org/conference/usenixsecurity22/presentation/arp

[53] Y. Yao, J. Duan, K. Xu, Y. Cai, Z. Sun, and Y. Zhang, "A survey on large language model (LLM) security and privacy: The good, the bad, and the ugly," *arXiv preprint arXiv:2312.02003v3*, 2024. [Online]. Available: https://arxiv.org/abs/2312.02003v3

[54] A. Shahriar, M. N. Rahman, S. Ahmed, F. Sadeque, and M. R. Parvez, "A survey on agentic security: Applications, threats and defenses," *arXiv preprint arXiv:2510.06445v1*, 2025. [Online]. Available: https://arxiv.org/abs/2510.06445v1

[55] M. A. Ferrag, N. Tihanyi, D. Hamouda, L. Maglaras, and M. Debbah, "From prompt injections to protocol exploits: Threats in LLM-powered AI agents workflows," *arXiv preprint arXiv:2506.23260v1*, 2025. [Online]. Available: https://arxiv.org/abs/2506.23260v1

[56] LangChain, "LangGraph, version 1.0.3," Software release, November 10, 2025. [Online]. Available: https://pypi.org/project/langgraph/1.0.3/

[57] E. Debenedetti, I. Shumailov, T. Fan, J. Hayes, N. Carlini, D. Fabian, C. Kern, C. Shi, A. Terzis, and F. Tramèr, "Defeating prompt injections by design," 2025, version 2. [Online]. Available: https://arxiv.org/abs/2503.18813v2

[58] M. Costa, B. Köpf, A. Kolluri, A. Paverd, M. Russinovich, A. Salem, S. Tople, L. Wutschitz, and S. Zanella-Béguelin, "Securing AI agents with information-flow control," 2025, version 2. [Online]. Available: https://arxiv.org/abs/2505.23643v2

[59] T. Shi, J. He, Z. Wang, H. Li, L. Wu, W. Guo, and D. Song, "Progent: Programmable privilege control for LLM agents," 2025, version 2. [Online]. Available: https://arxiv.org/abs/2504.11703v2

[60] S. Yao, N. Shinn, P. Razavi, and K. Narasimhan, "τ-bench: A benchmark for tool-agent-user interaction in real-world domains," 2024, version 1. [Online]. Available: https://arxiv.org/abs/2406.12045v1

[61] E. Debenedetti, J. Zhang, M. Balunović, L. Beurer-Kellner, M. Fischer, and F. Tramèr, "AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents," in *Advances in Neural Information Processing Systems*, vol. 37, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/97091a5177d8dc64b1da8bf3e1f6fb54-Abstract-Datasets_and_Benchmarks_Track.html